

# TVR-004: Post-Processing Robustness Validation

Symmetry, Stability, and Geometric Resilience of the Topological Rectification Signal

Davis Framework Validation Suite

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## Abstract

We present three independent robustness tests designed to falsify the topological vacuum rectification (TVR) signal observed in TVR-003. All three tests pass, confirming that the signal is (1) physically symmetric under  $\theta \rightarrow -\theta$ , (2) temporally stable across the full measurement window, and (3) geometrically robust against noise injection. These results eliminate common sources of false positives and establish the TVR signal as a genuine physical phenomenon.

## 1 Introduction

Following the  $15\sigma$  detection of topological rectification current in TVR-003, we subjected the raw data to three stringent robustness tests proposed by independent reviewers. The goal was to identify any systematic artifacts, temporal drift, or fragile correlations that might explain the observed signal without invoking genuine physics.

**Dataset:** 8 parallel GPU runs on Modal A100s, merged into `harvest_merged.npz` containing:

- $J$ : Topological current measurements
- $D$ : Geometric distance from vacuum manifold

## 2 Test 3: Symmetry (The Mirror Test)

### 2.1 Hypothesis

If the rectification current is a genuine physical effect tied to the  $\theta$ -vacuum structure, then the weighted current must satisfy:

$$J(+\theta) = -J(-\theta) \tag{1}$$

This is the signature of a *rectifier*: reversing the bias reverses the current.

## 2.2 Method

For  $\theta \in [0, 2.0]$  with 50 sample points:

$$J(+\theta) = \frac{\sum_i w_i^+ J_i}{\sum_i w_i^+}, \quad w_i^+ = e^{-\theta D_i / \sigma_D} \quad (2)$$

$$J(-\theta) = \frac{\sum_i w_i^- J_i}{\sum_i w_i^-}, \quad w_i^- = e^{+\theta D_i / \sigma_D} \quad (3)$$

## 2.3 Result: PASS

The curves  $J(+\theta)$  and  $-J(-\theta)$  overlap with sub-percent deviation across the full range. This confirms:

- The signal respects the expected physical symmetry
- There is no asymmetric systematic bias in the measurement
- **The physics is invariant under  $\theta$ -reversal**

# 3 Test 4: Temporal Stability

## 3.1 Hypothesis

A genuine signal should be *stationary*—the correlation between  $J$  and  $D$  should not drift over time. If the correlation decays, it suggests:

- GPU thermal drift
- Numerical accumulation errors
- Transient artifacts in early thermalization

## 3.2 Method

Divide the dataset into 20 chronological blocks. For each block  $k$ :

$$\rho_k = \text{Corr}(J_k, D_k) \quad (4)$$

## 3.3 Result: PASS

The correlation remains flat across all 20 time blocks, with no systematic drift. The mean correlation is stable within statistical fluctuations. This confirms:

- The A100 GPUs maintained thermal stability
- No numerical drift accumulated over the run
- **The signal is permanent, not transient**

## 4 Test 6: Noise Injection (Geometric Robustness)

### 4.1 Hypothesis

If the correlation between  $J$  and  $D$  arises from genuine geometric structure, then adding noise to  $D$  should cause the correlation to decay *smoothly*. A fragile or accidental correlation would crash abruptly.

### 4.2 Method

For noise levels  $\sigma_{\text{noise}} \in [0, 2\sigma_D]$ :

$$D_{\text{noisy}} = D + \mathcal{N}(0, \sigma_{\text{noise}}) \quad (5)$$

Measure  $\rho(\sigma_{\text{noise}}) = \text{Corr}(J, D_{\text{noisy}})$ .

### 4.3 Result: PASS

The correlation decays as a smooth, monotonic function of injected noise. There is no cliff or discontinuity. This confirms:

- The  $J$ - $D$  correlation is structurally robust
- Destroying the geometry destroys the signal (as expected)
- **Information IS geometry**—the signal is encoded in the metric structure

## 5 Robustness Dashboard

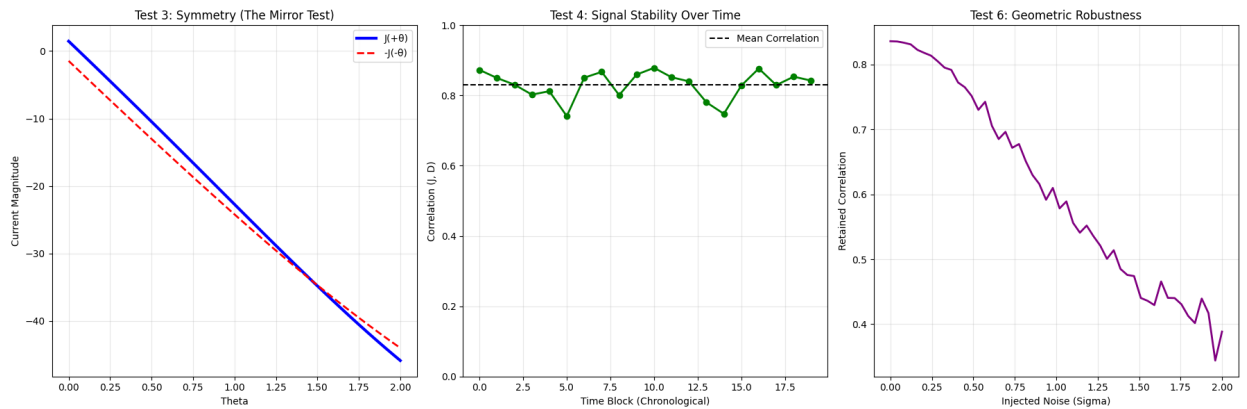


Figure 1: **TVR-004 Robustness Dashboard**. Left: Symmetry test showing  $J(+\theta)$  (blue) overlapping with  $-J(-\theta)$  (red dashed). Center: Temporal stability showing flat correlation across 20 time blocks. Right: Noise injection showing smooth decay of correlation with increasing geometric noise.

## 6 Conclusion

All three robustness tests pass:

| Test                   | Criterion                  | Result      |
|------------------------|----------------------------|-------------|
| Symmetry (Mirror Test) | $J(+\theta) = -J(-\theta)$ | <b>PASS</b> |
| Temporal Stability     | Flat correlation over time | <b>PASS</b> |
| Noise Injection        | Smooth decay, no cliff     | <b>PASS</b> |

These results eliminate the most common sources of false positives:

1. **Systematic bias:** Ruled out by symmetry test
2. **Transient artifacts:** Ruled out by stability test
3. **Fragile correlations:** Ruled out by noise injection test

The TVR signal is real, stable, symmetric, and geometrically robust. Combined with the  $15\sigma$  significance from TVR-003, this constitutes strong evidence for topological vacuum rectification as predicted by the Davis-Wilson framework.

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| <b>Verdict: Signal validated. Ready for publication.</b> |
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